

Data Development Life Cycle :-

Establishing
Requirement

Data Requirement

data analysis

Conceptual
Data model

database design

Logical
Schema

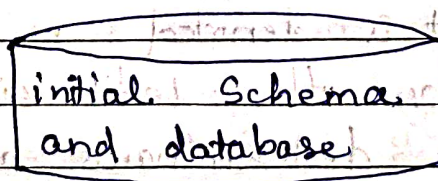
Implementation

initial Schema
and database

testing

Released schema
and database

maintenance



Relational Database Design:-

Functional Dependency:-

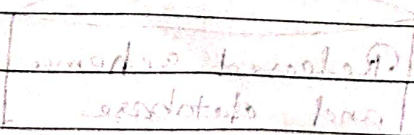
→ A Functional dependency occurs when 1 attribute uniquely determines another attribute within a relation. It is a constraint that describes how attributes in a table related to each other.

→ If attribute "A" is functionally determines attribute "B". $A \rightarrow B$

→ Functionally dependency are used to mathematically express the relation among database entities and are very important to understanding advanced concept in database system.

→ In relational database management system functional dependency is a concept that specifies the relationship between two set of attributes where one attribute determines the value of another attributes. It is denoted as $A \rightarrow B$ where the attribute set on the left side of the arrow A is called determinant and B is called dependent.

$A \rightarrow B$:- The value of A set should be present in set B.



Armstrong Properties of Functional Dependency:-

- (i) Reflexibility:- If B is a subset of A , then $A \rightarrow B$ holds by reflexivity rules.
- (ii) Argumentation:- If $A \rightarrow B$ is a valid dependency then $AC \rightarrow BC$ is also valid by the argumentation rule, where C is an attribute.
- (iii) Transitivity:- If $A \rightarrow B$, $B \rightarrow C$ i.e. $A \rightarrow C$.

Types of Functional Dependency:-

These are 4 types of FD in DBMS.

- (1) Trivial FD
- (2) Non-trivial FD
- (3) Multivalued FD
- (4) Transitivity FD

Trivial Functional Dependency:-

In trivial FD, a dependent always a subset of determinant, i.e. $x \rightarrow y$ and y is the subset of x then it is called trivial functional dependency.

Non-Trivial Functional Dependency:-

In non-trivial functional dependency the dependent is straightly not a subset of determinant, i.e. $x \rightarrow y$, y is not a subset of x , that it is called non-trivial FD.

Transitive FD:-

In transitive functional dependency dependent is indirectly dependent on the determinant, i.e. $A \rightarrow B$, $B \rightarrow C$ then according to the transitive Dependency $A \rightarrow C$. This is called as transitive functional dependency.

Fully Functional dependency:-

In fully functional dependency an attribute or a set of attribute uniquely determine another set of attribute.

$$X \rightarrow Y, Y \rightarrow Z, Z \rightarrow W, Z \rightarrow V.$$

Partial Functional Dependency:-

In partial functional dependency an attribute or a set of attribute uniquely determine another set of attribute.

In partial functional dependency a nonkey attribute depends on a part of the composite key rather than the whole keys.

If a relation R has attributes x, y, z where x & y are the composite key and z is non key attribute, then $x \rightarrow z$ is a partial functional in RDBMS.

Functional dependency have numeric application in the field of DBMS. Here some of the application are

- (i) Data Normalization
- (ii) Query Optimization
- (iii) Consistency of data
- (iv) Data Quality improvement.

Rules in Functional Dependency:-

- (1) Reflexive
- (2) Augmentation
- (3) Transitive

Normalization :-

→ Process to improve data integrity and reduce data redundancy.

→ Various types of normal form are

- (1) 1NF
- (2) 2NF
- (3) 3NF
- (4) BCNF
- (5) 4NF
- (6) 5NF

→ Normalization is a process of organizing the data in database to avoid data redundancy, setting anomalies, updated anomalies and its deletion anomalies. A process of anomaly (insert, update, delete) in such a way that the results coming to the database are always unambiguous and is maintained.

search normalization interesting to relational database theory. It may have the effort of duplicating data within the database and obtain results in the setting of additional tables, i.e. called as normalization.

→ Database normalization is the process of removing the data from your tables into improve storage efficiency, data integrity and scalability.

→ Normalization generally involves splitting the existing tables into multiple tables, which must be rejoined or linked each time a query is issued. That's why normalization otherwise called as broken table process.

1NF:- (First normal form)

→ A relation said to be in first normal form if it contain a atomic value.

→ 1NF states that an attribute of a table can not hold multiple values. It must hold only single valued attribute.

→ 1NF disallows the multivalued attributes, composite attributes, composite key and their combinations.

Std

Sl. NO	S-Name	S-Sub
01	Siba	C, C++, Java
02	Rama	C, DBMS, OS
03	Hazi	DBMS
04	Krishna	CPDS

Std (NF1)

Sl. NO	S-Name	S-Sub1	S-Sub2	S-Sub3
01	Siba	C	C++	Java
02	Rama	C	DBMS	OS
03	Hazi	DBMS	NULL	NULL
04	Krishna	CPDS	NULL	NULL

(Single valued from column)

Std (NF1)

Sl. NO	S-Name	S-Sub
01	Siba	C
02	Siba	C++
03	Siba	Java
04	Rama	C
05	Rama	DBMS
06	Rama	OS
07	Hazi	DBMS
08	Krishna	CPDS

(Single valued for row)

2NF:- (Second normal Form)

- In the 2NF relational must be in 1NF.
- In the 2NF all the non-key attributes are fully functional dependent on primary key.

(CK)	(PK)	(SK)	(com.k)
Sl.NO	Emp-Id	Emp-Name	Emp-add
01	Emp-01	Ram	BBSR, BAM
02	Emp-02	Sita	CTC
03	Emp-03	Siba	BAM, BBSR
04	Emp-04	Laumi	BBSR

(CK)	(PK)	(SK)		
Sl.NO	Emp-Id	Emp-Name	Emp-add1	Emp-add2
01	Emp-01	Ram	BBSR	BAM
02	Emp-02	Sita	CTC	NULL
03	Emp-03	Siba	BAM	BBSR
04	Emp-04	Laumi	BBSR	NULL

- Here Emp-add1 & Emp-add2 are non key attribute.

(PK)	(Non key Attribute)
Emp-Id	Emp-add2
Emp-01	BAM
Emp-02	NULL
Emp-03	BBSR
Emp-04	NULL

3NF:- (Third normal form)

- A relation will be in 3NF if it is in 2NF and not contain any transitive partial dependency.
- 3NF is used to reduce the data duplication.
- 3NF is also used to achieve the data integrity.
- If there is non transitive dependency for non-prime attributes, then the relation must be in 3NF.
- IMP → A relation is in 3NF if it holds atleast one of the following attributes, then the relation is in 3NF conditions for every non-trivial dependency $x \rightarrow y$
 - (i) X is a superkey
 - (ii) Y is a prime attribute, i.e. each element of Y is part of some candidate key.

Sl.No	S_Name	S_ID	S_Add	S_Ph.No.
1	Ram	M-01	BBSR	9981, 9111
2	Hari	M-02	BDR	9821, 9222
3	Sita	M-03	BLSR	9342, 9333
4	Laxmi	M-04	BAM	9671, 9731

S_ID	S_Ph.No.	S_Name
M-01	9981	Ram
M-02	9821	Hari
M-03	9342	Sita
M-04	9641	Laxmi

S-ID

S-addr

M-01

BBSB

M-02

BDK

M-03

BLSB

M-04

BAM

BCNF :-

- BCNF stands for Boyce-codd normal form
- BCNF is the advanced version of 3NF
- BCNF is stricter than 3NF
- A table is in BCNF if every functional dependency, $u \rightarrow v$, u is the super key of the table.
- For BCNF that table should be in 3NF and for every functional dependency, L.H.S is superkey

Sl. NO	S-ID	S-Name	Addr	Ph. NO.
01	001	Ram	BBSB	12345
02	002	Sita	CTC	67891
03	003	Sita	BAM	12456
04	004	Krishna	PURI	78912
(u)	(PK)	(SK)	(Non-u)	(Comp. key)

Finding the SK:-

S_Name S-Id

Ram 001

Siba 002

Sita 003

Krushna 004

→ BCNF is an advanced version of 3NF that's why it is also referred as 3.5 NF. BCNF is stricter than 3NF and a table complies with BCNF.

(i) K is superkey.

(ii) For every functional dependency K dependent to Y (Non prime attributes are functionally dependent on not prime attribute.)

BCNF Table

S_Name	S-Id	Add
Ram	001	DBSR
Siba	002	CTC
Sita	003	BADM
Krushna	004	Puri

S_Name	S-NO.	Add
Ram	001	
Siba	002	
Sita	003	
Krushna	004	

4NF :- (Fourth normal form)

→ A relation will be in 4NF if it is in BCNF and has no multivalued dependency.

→ For a dependency $A \twoheadrightarrow B$, if for a single value of A, multivalues of B exists, then the relation will be a multivalued dependency.

S_Id	Course	Hobby	Name	Ph.No
001	MCA	Playing	Ram	1234
002	MBA	Singing	Hari	2345
003	MCA	Dancing	Sita	5678
004	MBA	Travelling	Siba	8912

S_Id	Hobby	S_Id	Course
001	Playing	001	MCA
002	Singing	002	MBA
003	Dancing	003	MCA
004	Travelling	004	MBA

S_Id	Name
001	Ram
002	Hari
003	Sita
004	Siba

5NF:- (Fifth normal form)

→ A relation is in 5NF if it is in 4NF and not contains any join dependency and joining should be lossless.

→ 5NF is satisfied when all the tables are broken into as many tables is possible in order to avoid the redundancy.

→ 5NF is also known as project - join - normal form (PJ/NF).

<u>S_Name</u>	<u>add</u>	<u>Hubby</u>	<u>BF</u>
Ram	BBSR	Singing	Bisnu
Hari	CTC	Dancing	Nikil
Seta	BAM	Playing	Sahil
Seba	PUR	Travelling	Rohan

Relational Decomposition:-

When a relation in the relational model is not in appropriate Normal form then the decomposition of a relation is required.

→ In a database it breaks the table into multiple tables.

→ If the relation has no proper decomposition then it may lead to problems like "loss of information".

→ Decomposition is used to eliminate some of the problems of bad design like anomalies, inconsistency and redundancy.

→ It is of 2 types

- (1) Lossless decomposition
- (2) Dependency preserving

Lossless Decomposition :-

→ If the information is not lost from the relation i.e. decomposed then the decomposition will be lossless.

→ The relation is said to be lossless decomposition if natural joins of all the decomposition give the original relation.

Join dependency / SQL Join :-

As the name shows 'join' means to combine something. In case of SQL join means combine two or more tables.

→ In SQL join is used to combine the records from 2 more tables.

Types of SQL Join :- Inner Join

Left Join outer join

Right outer join

Full outer join

Conditional Join :-

It is a special type of join in which we join more than 2 relations and the join is governed by some condition or some condition.

Natural Join :-

It is a special type of join in which the equality condition is true for all attributes of the relation.

Left Outer Join :-

It is a special type of join in which the join considered as all the element of left relations and the tuple from right relations that satisfied the given condition is called as left outer join.

Right Outer Join :-

Right outer join is special type of join in which the join conditions, all the element of right relation and the tuple from left relation are present in a relation.